SMART QUEUE MANAGEMENT SYSTEM FOR UNIVERSITY CLINICS

Concept Paper — Revised (version 2)
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Revisions in this version: the problem statement was narrowed to the setting of the study, and the objectives were rewritten to be measurable, as asked by the Instructor 1 and the Adviser on the first version.

Background of the Study

University clinics serve a steady stream of students, faculty and staff who arrive for consultations, medical clearances, dental check-ups and the release of laboratory results. At most campuses the arrival of these patients is still managed by a paper logbook at the reception desk and a verbal call of names once a nurse or physician is free.

Digital queue management systems address this by issuing a numbered ticket, displaying the current number, and notifying the patient when their turn approaches. Commercial systems are priced and designed for large institutions; a lightweight web-based system running on the phones students already carry would suit a campus clinic better.

Statement of the Problem

Patients cannot see their position in line, staff are interrupted by repeated questions, urgent cases wait behind routine clearances, and the clinic cannot report service volume.

Objectives

The study aims to develop and evaluate a web-based Smart Queue Management System for the university clinic. Specifically, it aims to:

1. analyze the current patient reception and queueing process of the clinic and identify its problems;

2. design and develop a system that lets patients join the queue through a QR code or the university portal, view their position and estimated waiting time, and receive an email when their turn approaches;
3. provide clinic staff with a dashboard to call, skip, recall and prioritize tickets, and to generate daily and monthly service reports;
4. evaluate the system in terms of functional suitability, performance efficiency, usability and reliability based on ISO/IEC 25010; and
5. measure the perceived usability of the system using the System Usability Scale.

Scope and Delimitations

The system covers the queue for walk-in services of the university clinic: medical consultation, dental consultation, medical clearance and the release of results. Patients sign in with their university email address; a patient may hold one active ticket at a time.

The study does not cover appointment scheduling, electronic medical records, prescriptions, billing or any clinical data beyond the service requested. Notifications are sent by email only; SMS is outside the scope because of its per-message cost.

Proposed Method

The group will gather requirements from the intended users, develop the system in two-week Scrum sprints, and evaluate it with ISO/IEC 25010 and the System Usability Scale.

Expected Output

Smart Queue Management System for University Clinics and Evaluation results.

References

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Dummy concept paper for testing the HAU-SOC prototype. All content is placeholder.